



# Pragati Engineering College

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# Workforce Participation Visualizer

## About the Project

The **Workforce Participation Visualizer** is a socially impactful C++ project designed to analyze and visualize workforce demographic data related to gender equality. The project focuses on identifying the participation levels of men and women across different employment sectors using CSV data processing and ASCII-based graphical representation.

In many industries around the world, gender imbalance still exists. Some sectors have significantly higher male participation, while others have higher female participation. This project aims to represent such disparities in a simple, understandable, and visually engaging way using programming concepts.

The system reads workforce statistics from a CSV file, processes the information, and generates colorful ASCII charts in the terminal window. These charts make it easier to compare gender participation percentages and observe workforce gaps instantly.

This project not only demonstrates technical concepts such as:

- File handling
- Data parsing
- Structures
- Loops
- Functions
- Visualization logic
- but also spreads awareness about the importance of:
  - Equal opportunities
  - Inclusive workplaces
    - Women empowerment
    - Gender-balanced employment

## Main Aim of the Project

The main aim of this project is to:

- Analyze workforce demographic data

- Compare male and female participation rates
- Identify gender gaps in different sectors
- Present data visually using ASCII charts
- Promote awareness about gender equality through technology

### **Why This Project is Important ?**

Gender equality is one of the most important goals in modern society. Despite progress in education and employment, many industries still show unequal workforce representation.

This project helps users:

- Understand workforce inequality easily
- Analyze participation trends sector-wise
- Learn data visualization basics
- Combine social awareness with programming skills

The visual charts generated by the program make complex numerical data more understandable even for beginners.

### **Project Workflow :**

#### **Step 1 — Read CSV File**

The system loads workforce demographic data from a CSV file.

#### **Step 2 — Extract Data**

The program separates:

- Sector Name
- Male Percentage
- Female Percentage

#### **Step 3 — Store Data**

Information is stored using structures and vectors.

#### **Step 4 — Generate ASCII Charts**

Bars are displayed using repeated block characters.

#### **Step 5 — Calculate Gender Gap**

The difference between male and female participation is calculated automatically.

### **Step 6 — Display Final Report**

The terminal shows a colorful formatted report.

### **Concepts Used in C++**

This project uses multiple important C++ concepts:

- Structures (struct)
- Functions
- Vectors
- Loops
- Conditional statements
- File handling (ifstream)
- String stream processing (stringstream)
- Formatting (iomanip)
- ANSI escape codes

### **Conclusion:**

The **Workforce Participation Visualizer** is an innovative and meaningful C++ project that combines data visualization with social awareness. By converting workforce demographic statistics into colorful ASCII charts, the system provides a simple yet effective way to understand gender participation gaps across industries.